Erik Mauricio

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Education

Cornell University Ithaca, NY

Bachelor of Arts in Computer Science, Minor in Information Science – GPA: 3.47/4.0 *Expected May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Information Retrieval, Artificial Intelligence, Full-Stack Development, Discrete Structures, Linear Algebra

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL (PostgreSQL, MSSQL), NoSQL (MongoDB), HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Express.js, Flask, PyTest, NumPy, Tailwind CSS
Tools & Technologies: Git, Docker, AWS, REST APIs, WebSockets, RAG, Claude Code, OpenAI API

Experience

County of Orange Santa Ana, CA

Data Engineer Intern *May 2026 – Aug. 2026*

- Translated requirements from the data analytics team into a full-stack Next.js data intake app with SSO and input validation, eliminating 60+ hours of annual Excel cleanup across 100 performance measures and 22 departments.
- Designed a normalized MSSQL database to centralize department KPIs and submission records, consolidating scattered Excel files into a single source of truth for reporting and data updates.
- Integrated MSSQL with Power BI DirectQuery to support real-time reporting for a public-facing budget dashboard serving 3M residents, replacing manual Excel pipelines with automated data updates.

Cornell Hack4Impact Ithaca, NY

Software Developer *Aug. 2025 – Present*

- Collaborate on an Agile, cross-functional team to build full-stack software that modernizes nonprofit workflows.
- Developed REST API endpoints and interactive Next.js map components for Rethink Food's nonprofit food-distribution platform, including sidebars, detail popups, and relationship views for 204 partner organizations.
- Integrated Mapbox and Geocoding APIs to render color-coded locations for 135 CBOs and 69 restaurants in NYC.
- Engineered a Deck.gl network visualization with curved paths to model food distribution across restaurant-CBO relationships serving 30K meals per week.

Cornell University Ithaca, NY

Web Development Undergraduate Teaching Assistant *Aug. 2025 – Present*

- Lead weekly labs and office hours across 3 terms for 40+ students, supporting full-stack exercises and projects.

Projects

Sports Discovery Engine | GitHub | *Next.js, TypeScript, Python, PyTest, BM25, SVD/LSA, Information Retrieval*

- Developed a full-stack sports team discovery web app with Next.js and Python that interprets natural-language queries, preferences, filters, comparisons, and exclusions to recommend relevant teams.
- Implemented a hybrid information retrieval pipeline using BM25, TF-IDF, and SVD/LSA and evaluated it with a PyTest suite, achieving 24.6% higher NDCG@10 and 9.7% higher Recall@10 than the strongest lexical baseline.
- Designed a RAG pipeline with LLM-based query rewriting, feature-based reranking, and diversity-aware retrieval to generate more relevant recommendations with evidence-grounded explanations.

Tiempos Perfectos | GitHub | *React, Tailwind CSS, Python, FastAPI, SQLAlchemy, PostgreSQL, BKT*

- Built a full-stack adaptive English-learning app using React, FastAPI, PostgreSQL, and Python to identify learners' mistakes, model concept-level mastery, and recommend personalized practice exercises.
- Implemented Bayesian Knowledge Tracing to estimate learner mastery and reduced prediction log loss by 10% vs. default parameters and 6.3% vs. a global model.
- Engineered a real-time AI speaking role play exercise using FastAPI WebSockets that adapts interactive conversations to users' concept-level mastery and asynchronously analyzes responses.

Leadership

Corporate Chair | *Cornell Society of Hispanic Professional Engineers* May 2026 – Present

- Secured 4 corporate sponsorships and planned 3 recruiting and professional development events through outreach to 80+ companies.